Gated Multi-Head Latent Attention

Deep Dive into Kimi K3 — Chapter 6

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Foundation Books Project

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Goal: define the compact record, prove the absorbed execution exact, show where the K3 gate stops absorption, trace a reproducible layer, and keep cache, work, and timing in separate ledgers.

Roadmap

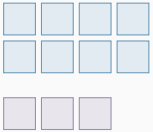
The compact record

Exact absorbed execution and the gate

Trace and ledgers

The compact record

One record per position, read by every head

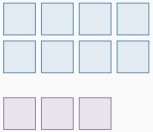


per-head K, V

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- Toy record width $2 + 1 = 3$; K3 $512 + 64 = 576$ — versus 30,720 materialized per position ($160/3\times$).

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- Toy record width $2 + 1 = 3$; K3 $512 + 64 = 576$ — versus 30,720 materialized per position ($160/3\times$).
- The shared branch counts: dropping it changes scores and understates the cache slope by $1/9$.

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Kernel fusion is fine; weight fusion is obstructed.

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Cache width alone does not predict runtime. Next: Chapter 7 mixes across depth.